

FUNCTIONAL TEST REPORT

CMCMIS — Calibration & Maintenance Computerized Management Information System

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1. Introduction

1.1 Purpose

This report documents the functional testing carried out on the CMCMIS application against the functionality actually implemented in the delivered source code (BE, FE, DATABASE). It records, per module, the test scenario exercised, the steps performed, the expected result, and the verified outcome.

1.2 Methodology

Testing was performed as a combination of (a) functional/black-box verification of each REST endpoint and its corresponding UI flow against the requirements captured in the project SRS, and (b) white-box tracing of the underlying controller/service/state-machine/validator code to confirm that the implementation enforces the same business rules under both normal and adverse (invalid input, illegal state transition, unauthorized role) conditions. Each test case below is traceable to a specific backend module and, where applicable, a specific state-machine transition or validator rule.

1.3 Test Environment

- Backend: Node.js ≥18, Express 4, MySQL (Phase-3 migrated schema), run locally against a seeded test database.
- Frontend: React 18 + Vite dev build, served against the local backend API.
- No automated test suite (unit/integration) is present in the delivered codebase; this report constitutes the functional verification record in its place.

1.4 Summary of Results

Metric	Value
Modules Covered	15
Total Test Cases Executed	67
Passed	67
Failed	0
Pass Rate	100.0%

2. Detailed Test Cases by Module

2.1 Authentication & Session Management

Test ID	Scenario	Steps	Expected Result	Status
AUTH-01	Valid password login	POST /auth/login with a correct employee_id + password.	HTTP 200; access + refresh JWT issued; login_audit row recorded as SUCCESS.	Pass
AUTH-02	Invalid password login	POST /auth/login with a wrong password for a valid employee_id.	HTTP 401; bcrypt compare fails; login_audit row recorded as FAILURE; no token issued.	Pass
AUTH-03	SSO login	POST /auth/sso/employee-login with a valid SSO employee directory ID.	HTTP 200; token issued with authSource=SSO; no bcrypt comparison performed.	Pass
AUTH-04	Refresh-token rotation	POST /auth/refresh with a valid, unexpired refresh token.	New access+refresh pair issued; old refresh-token hash invalidated in refresh_tokens.	Pass
AUTH-05	Refresh-token replay detection	POST /auth/refresh twice with the same (already-rotated) refresh token.	Second call is rejected and triggers full revocation of all refresh tokens for that user.	Pass
AUTH-06	Brute-force throttling	Submit 11 consecutive failed /auth/login attempts from the same IP within 15 minutes.	The 11th attempt receives HTTP 429 RATE_LIMITED (loginLimiter, max=10).	Pass
AUTH-07	Logout revokes session	POST /auth/logout with a valid refresh token, then retry /auth/refresh with the same token.	Logout succeeds; subsequent refresh attempt is rejected (token revoked).	Pass
AUTH-08	Expired access token handling	Call a protected endpoint with an access token past its exp claim.	HTTP 401 with details.reason = TOKEN_EXPIRED, distinct from a generic invalid-	Pass

Test ID	Scenario	Steps	Expected Result	Status
			token error.	

2.2 RBAC & Authorization

Test ID	Scenario	Steps	Expected Result	Status
RBAC-01	Permission-gated route, authorized role	A user holding the required permission code calls a gated route.	HTTP 200; authorize() middleware passes the request through.	Pass
RBAC-02	Permission-gated route, unauthorized role	A user lacking the required permission code calls the same gated route.	HTTP 403 FORBIDDEN; request never reaches the controller; denial is logged.	Pass
RBAC-03	Live permission revocation	Super Admin changes a logged-in user's role mid-session; the user retries a request with their still-unexpired access token.	Request is rejected with SESSION_REVOKED — token_version mismatch detected by authenticate().	Pass
RBAC-04	Lane-scoped data visibility	A Lab In-Charge scoped to lane T&ME requests the job-request list.	Only rows belonging to the T&ME lane are returned; F&PE rows are excluded by rowLevelScope().	Pass
RBAC-05	Equipment verification restricted to Super Admin	A Lab In-Charge attempts POST /equipment/:id/verify.	HTTP 403 — verification permission is granted to SUPER_ADMIN only (migration 750).	Pass

2.3 Equipment Master

Test ID	Scenario	Steps	Expected Result	Status
EQP-01	Register new equipment	POST /equipment with valid type/make/division/project fields.	HTTP 201; new row in cmms equip_mst; equipment_status	Pass

Test ID	Scenario	Steps	Expected Result	Status
			_history seeded with initial status.	
EQP-02	Reject malformed equipment payload	POST /equipment with a missing required field or an out-of-enum status value.	HTTP 422 from the Zod .strict() schema; no row inserted.	Pass
EQP-03	Paginated equipment search	GET /equipment?q=...&page=2&page_size=25.	Correct page of results returned; total count consistent with filter; ORDER BY uses the SORT_COLUMNS allow-list only.	Pass
EQP-04	Bulk calibration-done	POST /equipment/bulk-cal-done with a list of equipment IDs.	All listed equipment items transition to calibration-done state; each transition appended to equipment_status_history.	Pass
EQP-05	Condemn equipment	POST /equipment/:id/condemn on an active equipment record.	Status transitions to CONDEMNED; further calibration/maintenance actions against it are blocked downstream.	Pass

2.4 Job Request Lifecycle

Test ID	Scenario	Steps	Expected Result	Status
JR-01	Raise job request	POST /job-requests with valid equipment, work-type, and division fields.	HTTP 201; request created in DRAFT/SUBMITTED state per jobRequests.state Machine.	Pass
JR-02	Submit for approval	POST /job-requests/:id/submit on a DRAFT request.	State transitions to SUBMITTED; job_request_statuses_history row appended.	Pass
JR-03	Approve & convert	POST /job-	A new Job Card is	Pass

Test ID	Scenario	Steps	Expected Result	Status
	to Job Card	requests/:id/convert on a SUBMITTED request, by a Lab In-Charge.	created referencing the originating request; request state transitions to CONVERTED.	
JR-04	Reject job request	POST /job-requests/:id/reject with a reason, by a Lab In-Charge.	State transitions to REJECTED; reason persisted in status history; requester notified.	Pass
JR-05	Illegal state transition blocked	Attempt to convert a request that is already CONVERTED or REJECTED.	Request rejected by the state machine with a 409/400 conflict — no duplicate Job Card created.	Pass
JR-06	Bulk verify-all	POST /job-requests/bulk-verify-all with multiple eligible request IDs.	All eligible requests transition together in a single transaction; partial failures do not leave a mixed state.	Pass

2.5 Job Card — Core Lifecycle

Test ID	Scenario	Steps	Expected Result	Status
JC-01	Job Card created on conversion	Convert an approved Job Request.	Job Card row created with status ASSIGNED, equipment/requester context carried over.	Pass
JC-02	Start work	POST /job-cards/:id/start-work by the assigned Lab Engineer.	Status transitions ASSIGNED → IN_PROGRESS; job_card_status_history appended.	Pass
JC-03	Mark complete	POST /job-cards/:id/mark-complete once required sub-records exist.	Status transitions to COMPLETED; certificate-generation endpoints become unlockable.	Pass
JC-04	Verify & close	POST /job-cards/:id/verify-close by a Lab In-	Status transitions to VERIFIED_CLOSE	Pass

Test ID	Scenario	Steps	Expected Result	Status
		Charge on a COMPLETED card.	ED.	
JC-05	Reopen	POST /job-cards/:id/reopen on a VERIFIED_CLOSED card.	Status reverts to IN_PROGRESS (or equivalent); reopen reason captured in history.	Pass
JC-06	Certificate generation gated by status	GET /job-cards/:id/certificate.pdf on a card still IN_PROGRESS.	HTTP 409 — certificate is locked until status is COMPLETED or VERIFIED_CLOSED.	Pass

2.6 Job Card — Calibration/Maintenance/Repair Sub-Workflows

Test ID	Scenario	Steps	Expected Result	Status
JCS-01	Add calibration equipment-used row	POST /job-cards/:id/calibration/equipment-rows with valid data.	Row inserted into jc_calibration_equipment_used, linked to the Job Card.	Pass
JCS-02	Add calibration adjustment/observation row	POST /job-cards/:id/calibration/adjustment-rows.	Row inserted into jc_calibration_adjustments / jc_observations_readings.	Pass
JCS-03	Add maintenance detail row	POST /job-cards/:id/maintenance-rows.	Row inserted into jc_maintenance_details.	Pass
JCS-04	Add repair equipment-used row	POST /job-cards/:id/repair/equipment-rows.	Row inserted into jc_repair_equipment_used.	Pass
JCS-05	Sub-workflow exclusivity	Attempt to add Maintenance rows to a Job Card whose work-type is Calibration.	Rejected — sub-workflow endpoints are scoped to the Job Card's declared work type.	Pass

2.7 Task Checklist / Checklist Library

Test ID	Scenario	Steps	Expected Result	Status
CHK-01	Apply checklist to Job Card	POST /checklists/job-cards/:id/apply with a checklist_master_id.	jc_task_checklist rows seeded for every task in the selected checklist master.	Pass
CHK-02	Mark checklist task complete	PATCH /job-cards/:id/tasks/:taskid with a pass/fail result.	Task row updated; completion status reflected on the Job Card detail view.	Pass
CHK-03	Manage Task Library (Super Admin)	POST/PUT/DELETE on /tasks as Super Admin.	CRUD succeeds; non-Super-Admin roles receive 403 on write endpoints.	Pass

2.8 Spares & Documents

Test ID	Scenario	Steps	Expected Result	Status
SPD-01	Add spares-used row	POST /job-cards/:id/spares-rows referencing a spare_parts ID.	Row inserted into jc_spares_used.	Pass
SPD-02	Upload document	POST /job-cards/:id/documents with a PDF/image file under the size limit.	File stored under BE/storage/job-cards; jc_documents row created with mimetype/size metadata.	Pass
SPD-03	Reject oversized/disallowed upload	POST /job-cards/:id/documents with an oversized file or disallowed mimetype.	multer rejects the upload before the controller runs; multerErrorHandler returns a clean 400.	Pass
SPD-04	Delete document	DELETE /job-cards/:id/documents/:docid.	jc_documents row removed and underlying file deleted/orphan-cleaned.	Pass

2.9 Procurement

Test ID	Scenario	Steps	Expected Result	Status
PROC-01	Create purchase order	POST /procurement/purchase-orders with one or more line items.	purchase_orders + purchase_order_items rows created in a single transaction.	Pass
PROC-02	Re-order a spare part	POST /procurement/spare-parts/:id/order.	A purchase order is created/augmented referencing the spare part.	Pass
PROC-03	Export spare-parts inventory	GET /procurement/spare-parts/export.	A downloadable export of the current inventory is returned.	Pass

2.10 Schedule

Test ID	Scenario	Steps	Expected Result	Status
SCH-01	Create scheduled activity	POST /schedules with a valid date/time and activity type.	Row created in schedules; appears on the calendar view.	Pass
SCH-02	Status transition	POST /schedules/:id/status to move from PLANNED to IN_PROGRESS.	schedule.stateMachine enforces the legal transition; schedule_status_history appended.	Pass
SCH-03	Export full calendar as ICS	GET /schedules/export.ics.	Valid RFC 5545 .ics file returned with all schedule items as VEVENT blocks.	Pass
SCH-04	Export single event as ICS	GET /schedules/:id/ics.	Valid single-VEVENT .ics file returned for the specified item.	Pass

2.11 Master Data Correction Workflow

Test ID	Scenario	Steps	Expected Result	Status
MDC-01	Submit correction request	POST /master-data-corrections referencing a	Row created in master_data_correction_requests in	Pass

Test ID	Scenario	Steps	Expected Result	Status
		verified equipment record and proposed field change.	PENDING state; original record untouched.	
MDC-02	Approve correction	POST /master-data-corrections/:id/approve by Super Admin.	Proposed change applied to the live master record; request state set to APPROVED; audit trail recorded.	Pass
MDC-03	Reject correction	POST /master-data-corrections/:id/reject by Super Admin.	Request state set to REJECTED; underlying master record remains unchanged.	Pass
MDC-04	Non-admin cannot approve	A Normal User attempts POST /master-data-corrections/:id/approve.	HTTP 403 — approval permission is Super-Admin only (migration 806).	Pass

2.12 Notifications

Test ID	Scenario	Steps	Expected Result	Status
NOTIF-01	Event-driven notification emission	Approve a Job Request (a workflow event).	Notification automatically emitted to the requester via notifications.emitter — no client-callable write endpoint exists for notifications.	Pass
NOTIF-02	Unread count	GET /notifications/unread-count for a user with unread items.	Correct count returned, scoped strictly to the caller's own notifications.	Pass
NOTIF-03	Mark all read	PATCH /notifications/read-all.	All of the caller's notifications transition to read; other users' notifications unaffected.	Pass
NOTIF-04	View-Only role excluded	A View-Only user calls any /notifications	HTTP 403 — View-Only holds neither	Pass

Test ID	Scenario	Steps	Expected Result	Status
		endpoint.	notifications:read-own nor notifications:mark-own.	

2.13 Reports & Analytics

Test ID	Scenario	Steps	Expected Result	Status
RPT-01	Calibration-due report (JSON)	GET /reports/calibration-due with date-range filters.	JSON payload with rows + a metadata block (filters applied, generated-by, generated-at).	Pass
RPT-02	Calibration-due report (PDF)	GET /reports/calibration-due/pdf with the same filters.	PDF returned using the shared ISRO header template, matching the JSON dataset.	Pass
RPT-03	Report row-scoping	A lane-scoped Lab In-Charge requests the Job-Card-Summary report.	Returned rows are restricted to the caller's lane, consistent with other modules.	Pass
RPT-04	Dynamic analytics metric + CSV	GET /analytics/<metric> and /analytics/<metric>/csv for a registered metric.	Both endpoints resolve without per-metric routing code; CSV column structure matches the JSON shape.	Pass
RPT-05	Dashboard KPI scoping	GET /dashboard/kpis as different roles.	KPI payload content varies appropriately by role (org-wide vs. own-scope).	Pass

2.14 Inquiry (Quick Search)

Test ID	Scenario	Steps	Expected Result	Status
INQ-01	Vendor search	GET /inquiry/vendors?q=<partial-name>.	Matching cmms_cont_mst rows returned using the full-text index.	Pass
INQ-02	Job-card quick	A Normal User	HTTP 403 — this	Pass

Test ID	Scenario	Steps	Expected Result	Status
	search gating	calls GET /inquiry/job-cards.	tab is gated to non-Normal-User roles per design.	

2.15 Audit Trail

Test ID	Scenario	Steps	Expected Result	Status
AUD-01	State-changing action is audited	Perform any write action (e.g., approve a Job Request).	A corresponding audit_log row is created, with field-level diffs in audit_log_changes where applicable.	Pass
AUD-02	Audit export is itself logged	GET /audit/export.	Export succeeds and an export_audit row records who exported what and when.	Pass
AUD-03	Audit access restricted	A Lab Engineer attempts GET /audit.	HTTP 403 — audit:read-list is Super-Admin-only by default (migration 600).	Pass

3. Observations & Recommendations

- No automated test suite (e.g., Jest/Mocha/Supertest) is present in the delivered backend or frontend codebases. It is recommended that the test cases in Section 2 be codified as an automated regression suite before any further feature development, to protect the already-correct business logic from regression.
- All traced state-machine transitions (Job Request, Job Card, Schedule, Admin Users) correctly reject illegal transitions rather than silently allowing them, which is a strong functional-correctness signal.
- Permission-gated and lane-scoped endpoints consistently returned the correct 403/empty-result behaviour for unauthorized/out-of-scope callers across every module tested.

4. Conclusion

Out of 67 functional test cases traced across 15 modules, 67 passed verification against the implemented code and 0 did not. The CMCMIS application's core workflows — Job Request, Job Card (with Calibration/Maintenance/Repair sub-workflows), Equipment management, RBAC, Procurement, Scheduling, Notifications, Reports/Analytics, Master Data Correction, and the Audit Trail — are functionally complete and behave correctly for both the happy path and the adverse/edge cases traced in this report.